EXPERIMENT – 2

```
UW PICO 5.09

#include<stdio.h>                //preprocessor directive to include builtin header file
#define PI 3.14                 //PPD to define a symbolic constant or macro
int main(void)
{
        int radius;             //variable definition
        float area;
        printf("Enter the radius: ");   //printf() is builtin function to display output string
        scanf("%d", &radius);           //scanf() is BIF to get inputs
        area = PI*radius*radius;        //compute
        printf ("Area = %f\n", area);
        return 0;
}
```

SECTION 1 : SOURCE CODE

- This source code contains preprocessors directives (#include), macros (#define), and single – line comments.

```
manalithakar@Manalis-MacBook-Air ~ % gcc -save-temps area.c
manalithakar@Manalis-MacBook-Air ~ % ls
a.out         area       area.c     area.o     Desktop     Downloads   hello c    hello.cpp   Movies    Pictures
Applications  area.bc    area.i     area.s     Documents   hello       hello.c    Library     Music     Public
manalithakar@Manalis-MacBook-Air ~ % ./a.out
Enter the radius: 5
```

STAGES OF COMPILATION :

This source code will go through four different processing stages :

1. **Preprocessing stage**
   Converts source code area.c => into preprocessing source code *.i

2. **Compilation stage :**
   Converts preprocessing source code *.i => into assembly code *.s

3. **Assemling stage :**
   Converts assembly code *.s => into output file *.o

4. **Linking stage :**
   Converts output file *.o => into linked executable file a.out

SAVING THE FILE AND EXECUTING :

For saving all temporary files during the stage of compilation , we are using –save-temps flag with gcc .

```
manalithakar@Manalis-MacBook-Air ~ % gcc -save-temps area.c
manalithakar@Manalis-MacBook-Air ~ % ls
a.out           area            area.c          area.o          Desktop         Downloads       hello c         he
Applications    area.bc         area.i          area.s          Documents       hello           hello.c         Li
manalithakar@Manalis-MacBook-Air ~ % ./a.out
```

HERE'S THE EXPLANATION OF EACH GENERATED STAGE FILE :

- .i file (area.i) : cleans up the code by deleting extra comments and expanding shorts like #include and PI into their full actual values.
- .s file (area.s) : converts the C code into assembly code.
- .o file (area.o) : converts the assembly code into machine code (binary) which is not understable by humans.
- a.out file : links the binary code with standard built-in functions to produce the final output.

FINAL OUTPUT :

```
manalithakar@Manalis-MacBook-Air ~ % ./a.out
Enter the radius: 5
Area = 78.500000
manalithakar@Manalis-MacBook-Air ~ %
```